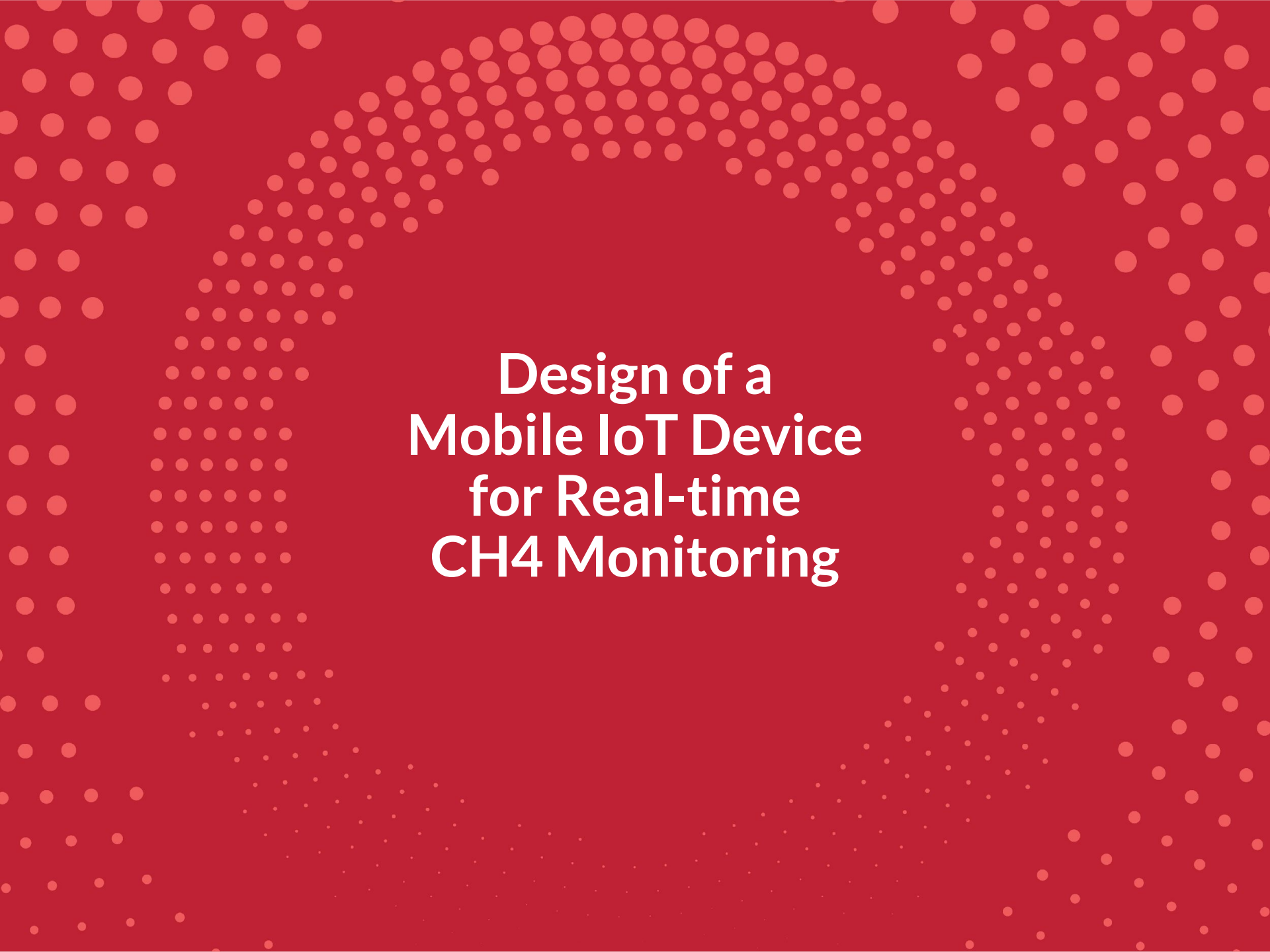




HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



Design of a Mobile IoT Device for Real-time CH₄ Monitoring



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

State of the art and on-site board design

Presentation of the project
State of the art
On-site board design

ONE LOVE. ONE FUTURE.



Presentation of the project

- Goals and targets
- Main checkpoints
- (project progress prevision
gantt diagram, ...)



Presentation of the project

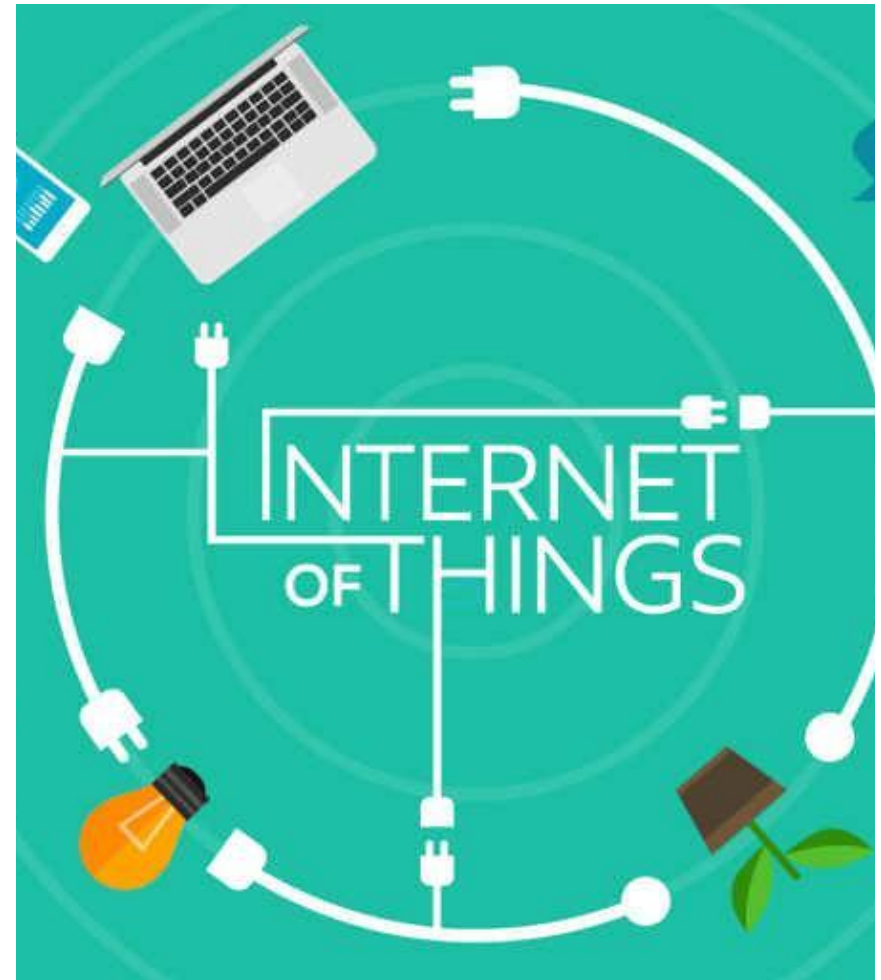
Design and development a mobile IoT device for real-time monitoring of methane gas levels

The device should be able to :

- **measure CH₄ concentrations**
- **transmit data for real-time monitoring and analysis**

This project targets :

- **agriculture applications**



Presentation of the project

Design a portable IoT device equipped with CH₄ gas sensors for accurate real-time monitoring.

Implement a user-friendly mobile application for real-time data visualization.

Develop firmware for data acquisition and wireless transmission to enable remote monitoring.

Ensure the device is power-efficient and capable of wireless connectivity for seamless data transfer and evaluation of performance

State of the art

- What is CH₄?
- Why focus on CH₄ instead of other gases?
- How can methane levels be measured?
- What level of precision is required in agriculture?
- What sensors are currently available on the market or being developed by researchers?

Introduction to Methane (CH₄)

Methane Sources:

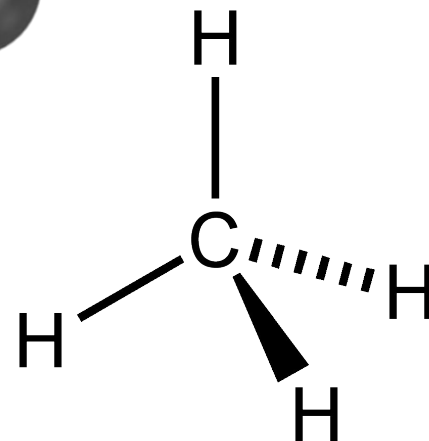
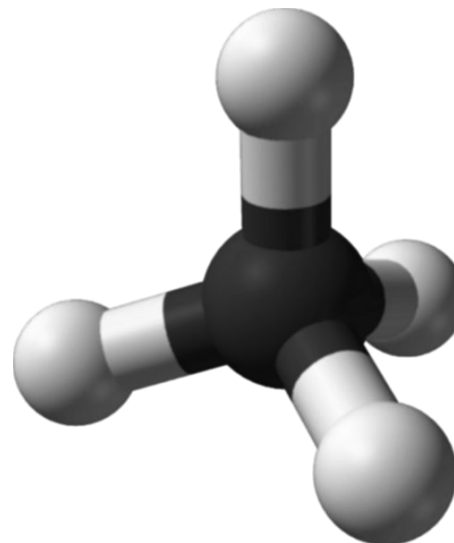
- Naturally present in soil, water, and the atmosphere
- Released by human activities (e.g., agriculture, rice paddies)
- Constitutes 60% of natural gas fossil deposits
- Produced by forced anaerobic fermentation

Uses:

- Electricity, heating, automotive fuel

Hazards:

- Highly flammable and explosive



Environmental Impact of Methane



Greenhouse Gas Contribution:

- Second most important greenhouse gas after water vapor
- Absorbs infrared radiation, contributing to global warming

Sources of Emissions:

- Anaerobic fermentation in wetlands, rice paddies, hydroelectric dams
- Fossil fuels

Methane Detection

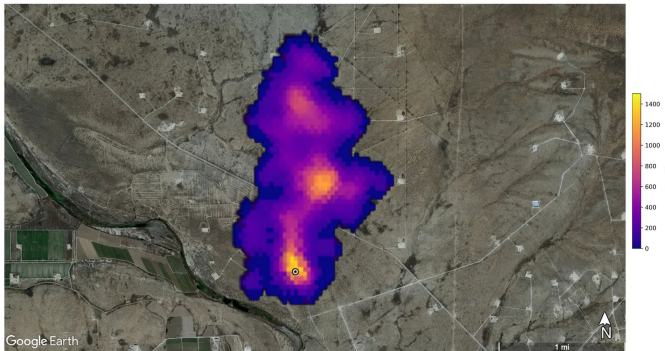


Detection Methods:

- Satellite or on-site sensors

Objective:

- Reducing methane emissions is more cost-effective and efficient for mitigating global warming compared to reducing CO₂



Mitigation Strategies and Gas Sensor Functionality

Strategies:

- Sensors + burning or distillation purification of CH₄ emissions
=> reduction of CO₂ emission
- Modify cattle diets
- Rice cultivation: reduce fertilizer use, combat water turbidity, and reduce methanogenic sediment production



Types of Gas Sensors:

- Electrochemical Sensors: Detect CO and SO₂
- Semiconductor Sensors: Sensitive to a wide range of gases
- Infrared Sensors: Detect CH₄ and CO₂

Market-Available Sensors

MQ4 Sensor:

- Detection range: 200-10000 ppm CH₄

MQ6 Sensor:

- Detection range: 300-10000 ppm (Propane and Methane)

MQ Sensor family :

- Semiconductor sensor
- Long range of sensitivity
- Low price
- Low consumption
- Unrestricted use condition (high range of temperatures resistance, high resistance from humidity, ...)



Tao	Using Tem	-10°C -50°C
Tas	Storage Tem	-20°C -70°C
R _H	Related humidity	less than 95%Rh

Detecting concentration
scope:
200-10000ppm
CH₄ , natural gas

Extract of MQ4 datasheet

Key Steps for Implementation

Embedded Board:

- Find a suitable methane sensor
- Use a microprocessor to acquire data
- Transmit data to the cloud, stored on platforms like ThingSpeak

User's Phone:

- Develop a mobile application to retrieve, process, and display data in real-time



On-site board design

- On-site Board Requirements
- Power Supply and Autonomy
- Component choices



On-site Board Requirements

Components:

- Methane sensor
- GSM/3G module
- Battery and autonomous battery charger
- STM32 microcontroller

Communication Options:

- Cellular device with SIM card (e.g., SIM800, SIM900)

Power Supply and Autonomy

Power Options:

- Battery-powered system for portability
- Solar-powered recharging system using an on-board solar cell

Environmental Constraints:

- Temperature range: 0°C to 50°C
- Humidity resistance: Up to 95%

Energy Autonomy:

- Renewable energies: solar and wind power

Communication module

Several devices are possible:

- a WiFi device, such as ESP8266 or ESP32
- a low-energy Bluetooth device, such as Nordic nRF52
- a LoRa device, such as RFM95
- **a cellular device => self-sufficient**

} not sufficient on their own
require a gateway

=> Not self-sufficient and can be battery powered module => PC and protocol in internet
BUT
Supply current required = 2 A >> Microcontroller
=> a separate DC/DC regulator



Microcontroller

Key points to decide :

- power consumption

- portability

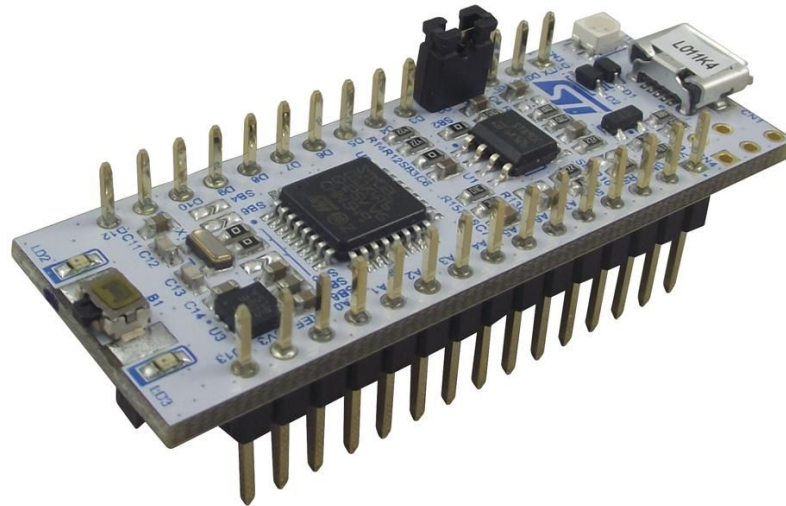
- resistance to environmental conditions

- board autonomy

Two categories fit :

STM32L0 and **STM32L4**

=> **Ultra-Low Power microcontrollers**



Nucleo-L011K4 with embedded STM32L011K4

Battery-powered system to ensure **portability**

System should be autonomous

=> **solar-powered** charging module with on-board solar cell

The STM32 Nucleo-32 board and its shield boards can be powered in three different ways from an external power supply, depending on the voltage used. The three power sources are summarized in the [Table 5](#).

Table 5. External power sources

Input power name	Connector pin	Voltage range	Max current	Limitation
VIN	CN4 pin 1	7 V to 12 V	800 mA	From 7 V to 12 V only and input current capability is linked to input voltage: 800 mA input current when $V_{IN}=7\text{ V}$ 450 mA input current when $7\text{ V} < V_{IN} < 9\text{ V}$ 300 mA input current when $10\text{ V} > V_{IN} > 9\text{ V}$ less than 300 mA input current when $V_{IN} > 10\text{ V}$
+5 V	CN4 pin 4	4.75 V to 5.25 V	500 mA	ST-LINK not powered
+3V3	CN4 pin 14	3 V to 3.6 V	-	ST-LINK not powered and SB14 and SB9 must be off.

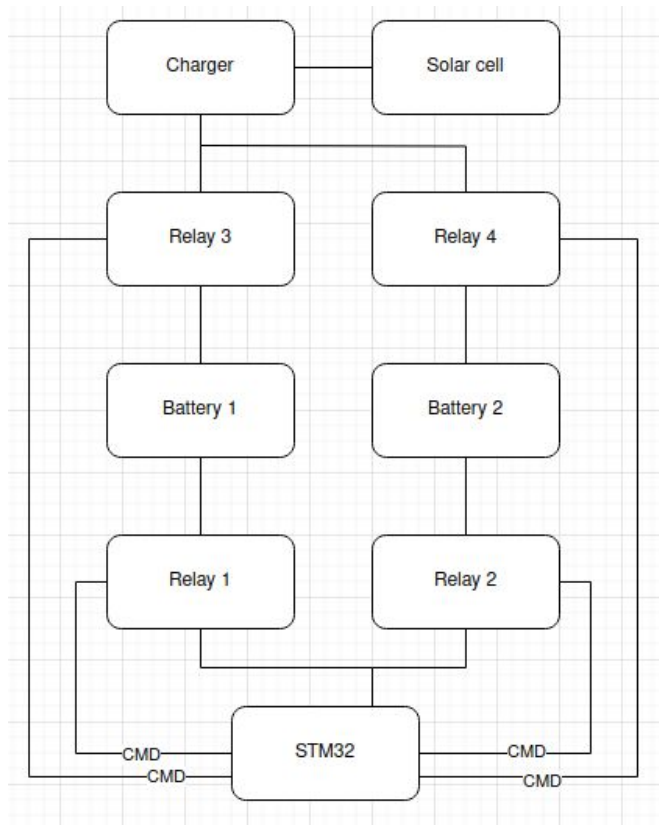


Battery Li-Ion 3.7V 2600mAh

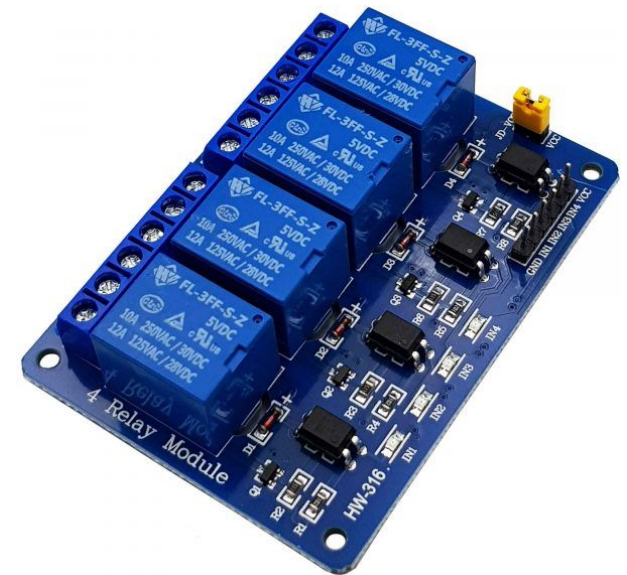
Extract of Nucleo-32 datasheet

Power supply

Autonomous system => two-batteries **relay** module

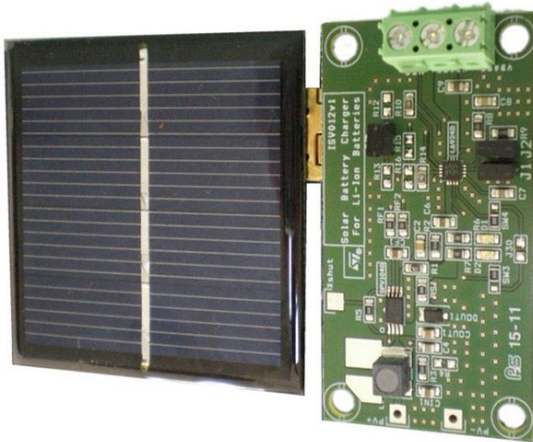


Theoric schematic
of charging and supply method



Module 4 Relay 5VDC

Charger and solar cell



**Solar Panel + Battery Charger
STEVAL-ISV012V1**

Includes features :
protect the battery against overcharging,
deep discharge,
and other potential problems

Summary:

- The project focuses on developing a methane sensor system for rice fields
- Emphasis on reducing methane emissions to mitigate global warming
- Combination of low-cost, sensitive sensors and renewable energy sources

Next Step:

- Design of a detailed schematic of the on-site system

A large, stylized graphic on the left side of the slide. It consists of a red background with a circular pattern of white dots of varying sizes, creating a sense of depth and movement. The word "HUST" is written in white, bold, sans-serif capital letters in the center of this graphic.

HUST

THANK YOU !